

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1712

Cavanaugh 138 kV
September 01, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1712. The Interconnection Customer proposes a 209.2 MW solar plus storage facility interconnecting at the Cavanaugh 138 kV substation in the Kentucky area. The request was received on April 05, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 516 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$217,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Springbank / Sta 11 138 kV Line	Barr Lake Substation / Kean Mtn And Grassy Creek 345 kV	105.6	397	5.9
Station 11 / Stock Island 138 kV L	Kean Mtn And Grassy Creek / Barr Lake 345 kV	105.4	416	5.9
Orovada / Carlton Hills 138 kV	Atoka / Carlton Hills 345 kV	103.0	329	8.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sta 11 / Springbank 138 kV L	Reconductor line	\$4,306,000	3% (\$143,000)
Sta 11 Substation / Stock Island 138 kV line	Reconductor line	\$1,746,000	3% (\$58,000)
Orovada / Carlton Hills 138 kV	Reconductor line	\$265,000	6% (\$16,000)
	Total allocated to project		\$217,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.